



Math Bootcamp

Module I: Linear Algebra, Part 2

Terrie Walmsley
Department of Economics
USCDornsife

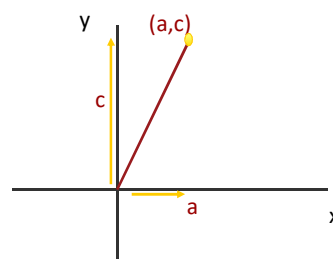


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Vectors and geometry

- $\mathbf{u} = \begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{n1} \end{bmatrix}$ is in $\mathbb{R}_n$ vector space $\mathbf{u} \in \begin{bmatrix} \mathbb{R} \\ \mathbb{R} \\ \dots \\ \mathbb{R} \end{bmatrix}$
 - So, $\mathbf{u} = \begin{bmatrix} a \\ c \end{bmatrix}$ is in $\mathbb{R}_2$ vector space
 - And any real number (a or c in $\mathbb{R}_2$ is in $\mathbb{R}$)
- Drawing a vector
 - a, c represent coordinates on the (x, y) plane
 - Line starts at origin and goes to point (a, c)
- Length of a vector:
 - $\|\mathbf{u}\| = [\sum_{i=1}^n a_i^2]^{1/2}$ (pythagoras theorem, Note that: $x^{1/2} = \sqrt{x}$)
 - $\|\mathbf{u}\| = [a^2 + c^2]^{1/2}$



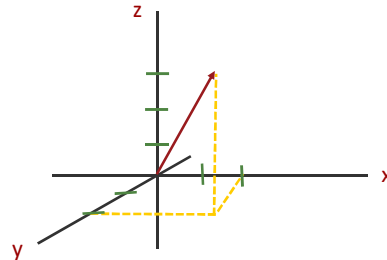
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Example: draw the vector and calculate its length

- $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

- Length of $\mathbf{u}$

- $\|\mathbf{u}\| = (2^2 + 2^2 + 3^2)^{1/2} = \sqrt[3]{17}$



Vector operations

- Dot product:

- $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \cdot \mathbf{v} = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

- Note that: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \cos \theta$

- Where θ is the angle between the two vectors.
 - If $\mathbf{u}$ and $\mathbf{v}$ are perpendicular the angle is 90 degrees.
 - Since $\cos 90 = 0$, then $\mathbf{u} \cdot \mathbf{v} = 0$ if $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.



Example



- Are $\mathbf{u}$ and $\mathbf{v}$ perpendicular (orthogonal)?

$$- \mathbf{u} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\bullet \mathbf{u} \cdot \mathbf{v} = 2 \times 1 + 2 \times 1 + 3 \times 2 = 8 \quad \text{Not perpendicular}$$

$$- \mathbf{u} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\bullet \mathbf{u} \cdot \mathbf{v} = 2 \times 1 + 2 \times 1 + 4 \times -1 = 0 \quad \text{Perpendicular}$$



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Basis for vector space



- $\mathbb{R}_n$ is defined by the minimum number of vectors N ($\mathbf{e}_1 \dots \mathbf{e}_N$) for which any vector in the space can be described as a linear combination of those vectors.
- E.g., the basis for $\mathbb{R}_2$ are:

$$- \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$- \mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3 \cdot \mathbf{e}_1 + 2 \cdot \mathbf{e}_2 = 3 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

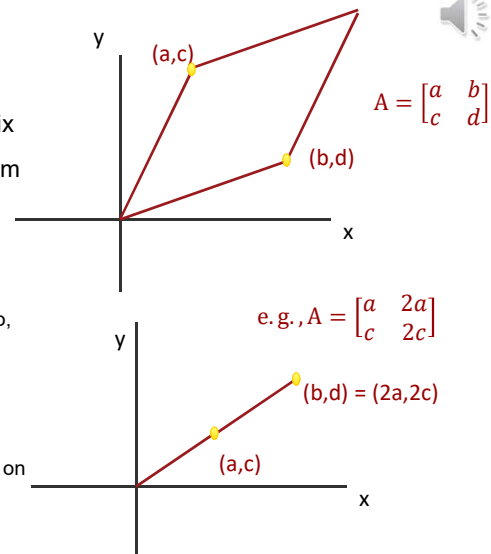


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Matricies and geometry

- Matrix can be drawn by drawing the vectors within the matrix
- Determinant of the matrix is the area under the parallelogram formed by the vectors within the matrix
 - Determinants can be negative - Negative area
 - If determinant of a 2x2 is zero then it is a straight line
 - 3x3 determinant is the volume of the square. If determinant is zero, then it has zero volume
 - For nxn matrix, if determinant = 0 then it loses a dimension (n-1)
- Linear independence
 - Matrix is linearly dependent if the determinant = 0 (e.g., vectors lie on same line in $\mathbb{R}_2$ or plane in $\mathbb{R}_3$)
 - Linearly independent if determinant is not equal to zero



Example

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}$$

Is this matrix linearly dependent?

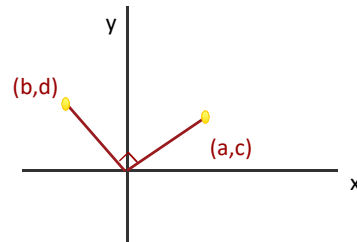
- $|A| = \sum_j^m a_{ij} \cdot C_{ij}$ no matter what i you choose
 - $i=1$ ($|A| = \sum_j^m a_{1j} \cdot C_{1j}$)
 - $1 \times (-2) + 3 \times (0) + 2 \times (1) = 0$
 - Yes, the matrix is linearly dependent



Orthogonal vectors and matrices



- Are two vectors orthogonal?
 - If the multiplication of two vectors equals zero ($u^T \cdot v = 0$) then they are orthogonal
- When is a matrix considered orthogonal?
 - Two Requirements
 - Each column (vector) is orthogonal ($u^T \cdot v = 0$)
 - Each column has length 1
 - If a matrix is orthogonal then the following is true:
 - $A \cdot A^T = I$
 - $A^T = A^{-1}$
 - Matrix A's transpose is also its inverse



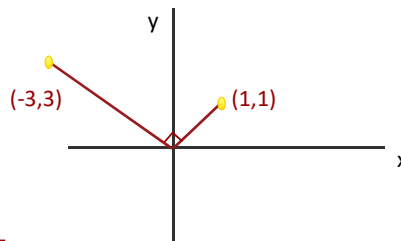
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Example: is the matrix orthogonal?



- $A = \begin{bmatrix} 1 & -3 \\ 1 & 3 \end{bmatrix}$
- If orthogonal
 1. $\begin{bmatrix} 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 3 \end{bmatrix} = 0$
 2. Lengths must equal 1: 1 and $\sqrt{18}$
- No, lengths of each vector must equal 1 for the matrix to be orthogonal, but they are perpendicular
- To make this matrix orthogonal, could divide all entries by the length



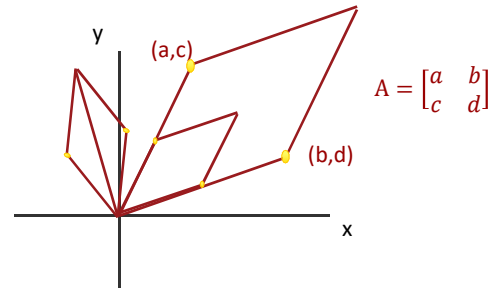
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Linear Transformation of Matrices/Vectors



- When we multiply or solve these vectors and matrices (equations) we change the matrices
 - Scaled
 - Rotated
 - May even drop a dimension $\mathbb{R}_n \rightarrow \mathbb{R}_{n-1}$ (e.g., $\mathbb{R}_3 \rightarrow \mathbb{R}_2$ means the plane becomes a straight line)



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Rank



- Rank is defined as the number of positive rows in row echelon form. Any row of zeros reduces the rank.
 - In an $n \times n$ matrix with rank of n (i.e., no zero rows, reduced row echelon form is the identity matrix), $|A| \neq 0$ and hence A^{-1} exists and the matrix is non-singular - there is a solution.
 - Column and row space is n , no null space
 - In an $n \times n$ matrix with rank of $< n$ (at least 1 zero row), $|A| = 0$ and hence A^{-1} does not exist and the matrix is singular and not invertible – there may not be a solution.
 - There is null space equal to the number of zero rows



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Exercise: find the Rank



Original matrices

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}$$

Reduced row echelon form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



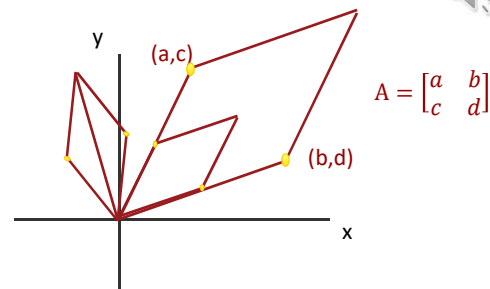
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Linear Transformations



- When we multiply or solve these vectors and matrices (equations) we change the matrices
 - Scaled
 - Rotated
 - May even drop a dimension (Rank)
- Be easier if we could put them in 90 degree space so they could not rotate.
- Need to find a new basis for vector space ($e_1 \dots e_N$) that ensures any vector multiplied by matrix A does not rotate
- How? Find eigen values and vectors such that:
 - $Av = \lambda v$



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Step 1: Finding the Eigen values (λ)



- To find the eigenvalues we solve for

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$(A - \lambda I) \cdot \mathbf{v} = 0$$

Prove this for
yourself

- Assuming $\mathbf{v}$ is non-singular and not equal to zero then that must mean that the determinant of $(A - \lambda I)$ is zero
 - Use $|A - \lambda I| = 0$ to solve for the eigenvalues λ



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Step 2: Obtain eigen vectors



- Find eigen vectors for **each** value of λ found
 - Solve $(A - \lambda I) \cdot \mathbf{v} = 0$ using λ 's to find two vectors $\mathbf{w}$ and $\mathbf{u}$



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Step 3: Create matrix Q



- Collect the λ 's and eigen vectors (**w** and **u**)
 - Make the eigen vectors length 1 (i.e., orthogonal), by dividing vectors by current length $\|v\|$ and $\|u\|$

$$\mathbf{w} = \frac{\mathbf{w}}{\|\mathbf{w}\|} \text{ and } \mathbf{u} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

- Create Q which is a matrix of the multiple eigen vectors of length 1, from previous step.

$$Q = \begin{bmatrix} \frac{\mathbf{w}}{\|\mathbf{w}\|} & \frac{\mathbf{u}}{\|\mathbf{u}\|} \end{bmatrix}$$

- Note that Q is orthogonal because $\mathbf{w}^T \cdot \mathbf{u} = 0$ and they have length 1.



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Example: Find eigen values



- $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$
- Eigenvalues of A
 - $|A - \lambda I| = 0$
 - $\begin{vmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = 0$
 - $(2-\lambda)^2 - 9 = 0$
 - $(2-\lambda) = \pm 3$
 - $-\lambda = -3 - 2$
 - $-\lambda = 3 - 2$
 - $\lambda = 5 \text{ or } -1$



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Example: Find eigen vectors using $(A - \lambda I)v = 0$

- For $\lambda = 5$
 - $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - 5 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - Reduced form $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - $u_1 = u_2$ (One equation and one unknown, set $u_1 = t$)
 - $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \cdot t \\ 1 \cdot t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t$
 - Eigen vector is: $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Check $A \cdot v = \lambda \cdot v$
 - $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- For $\lambda = -1$
 - $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - Reduced form $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - $u_1 = -u_2$ (One equation and one unknown, set $u_1 = t$)
 - $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \cdot t \\ -1 \cdot t \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot t$
 - Eigen vector is: $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- Check $A \cdot v = \lambda \cdot v$
 - $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$



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Example: Create Q Matrix

- $\lambda = 5$
- Vector: $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Calculate length:
 - $\|u\| = \sqrt[3]{1^2 + 1^2} = \sqrt[3]{2}$
- $\lambda = -1$
- $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- Calculate length:
 - $\|w\| = \sqrt[3]{1^2 + 1^2} = \sqrt[3]{2}$

$$Q = \begin{bmatrix} \frac{u}{\|u\|} & \frac{w}{\|w\|} \end{bmatrix} = \begin{bmatrix} 1/\sqrt[3]{2} & 1/\sqrt[3]{2} \\ 1/\sqrt[3]{2} & -1/\sqrt[3]{2} \end{bmatrix}$$



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Essence of A

- Note that:

$$Q.A.Q^T = \Lambda, \text{ where } \Lambda = \begin{bmatrix} \lambda \text{ from } u & 0 \\ 0 & \lambda \text{ from } w \end{bmatrix}$$

- This matrix Λ has the essentials of Matrix A (i.e., linear transformations of Λ have the same impact as linear transformations of A)

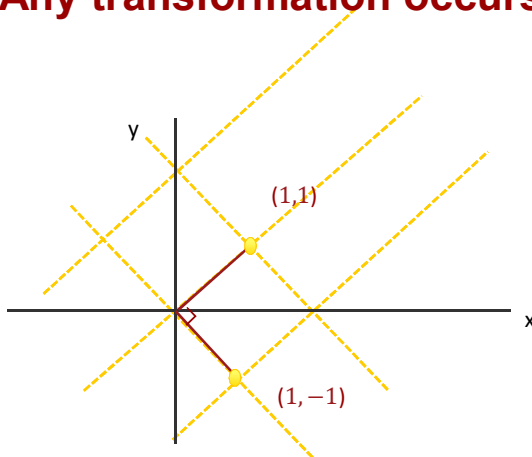
$$\bullet \Lambda = \begin{bmatrix} \lambda \text{ from } u & 0 \\ 0 & \lambda \text{ from } w \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$



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Any transformation occurs on this grid



- A matrix $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$
- Eigen vectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- $\lambda = 5$ and -1



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Helpful for powers?



- Great for powers

- $Q.A.Q^T = \Lambda$
- $Q^T.Q.A.Q^T.Q = I.A.I$
- $A = Q^T.\Lambda.Q$
- $A^2 = Q^T.\Lambda^2.Q$
- $A^n = Q^T.\Lambda^n.Q$

Since Q is orthogonal $Q^{-1} = Q^T$



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Trace: sum of diagonals



- Note also:

- $|A| = |Q|$
- Trace of A = trace of Q

- What is the Trace of A?

- Sum of diagonals: $\sum_{i=1}^n a_{ii} = a_{11} + a_{22} + a_{33}$

– e.g., $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}$

$$\text{Tr}(A) = 1+2+0=3$$

- Note: $\text{Tr}(A) = \text{Tr}(A^T)$



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Examples of Quadratics in matrix notation



- $g(x) = a \cdot x_1^2 + b \cdot x_2^2$
 $- g(x) = \mathbf{x}^T \cdot A \cdot \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \cdot \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
- $g(x) = x_1^2 + 4 \cdot x_1 \cdot x_2 + x_2^2$
 $- \mathbf{x}^T \cdot A \cdot \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
- $g(x) = a \cdot x_1^2 + b \cdot x_1 \cdot x_2 + c \cdot x_2^2$
 $- \mathbf{x}^T \cdot A \cdot \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \cdot \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$



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Put this quadratic in matrix form



- $g(x) = a \cdot x_1^2 + b \cdot x_2^2 + c \cdot x_3^2 + d \cdot x_1 \cdot x_2 + e \cdot x_1 \cdot x_3 + f \cdot x_2 \cdot x_3$
 $= \mathbf{x}^T \cdot A \cdot \mathbf{x}$
 $= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \cdot \begin{bmatrix} a & d/2 & e/2 \\ d/2 & b & f/2 \\ e/2 & f/2 & c \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
 $= \begin{bmatrix} a \cdot x_1 + \frac{d}{2} \cdot x_2 + \frac{e}{2} \cdot x_3 & \frac{d}{2} \cdot x_1 + b \cdot x_2 + \frac{f}{2} \cdot x_3 & \frac{e}{2} \cdot x_1 + \frac{f}{2} \cdot x_2 + c \cdot x_3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
 $= a \cdot x_1 \cdot x_1 + \frac{d}{2} \cdot x_2 \cdot x_1 + \frac{e}{2} \cdot x_3 \cdot x_1 + \frac{d}{2} \cdot x_1 \cdot x_2 + b \cdot x_2 \cdot x_2 + \frac{f}{2} \cdot x_3 \cdot x_2 + \frac{e}{2} \cdot x_1 \cdot x_3 + \frac{f}{2} \cdot x_2 \cdot x_3 + c \cdot x_3 \cdot x_3$
 $= a \cdot x_1^2 + b \cdot x_2^2 + c \cdot x_3^2 + d \cdot x_1 \cdot x_2 + e \cdot x_1 \cdot x_3 + f \cdot x_2 \cdot x_3$



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